

Linear/Matrix Algebra Review

26 Points Possible

Statistics 469: Analysis of Correlated Data

1. Compute the following in **R**:

- a. $\mathbf{A} + \mathbf{B}$
- b. $\mathbf{A} - \mathbf{B}$
- c. $\mathbf{AB}$
- d. $\mathbf{BA}$

where

$$\mathbf{A} = \begin{pmatrix} 8 & 6 \\ 6 & 8 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 1 & 8 \\ 3 & 5 \end{pmatrix}$$

2. If possible, compute the following in **R**. If non-conformable show that R throws an error.

- a. $\mathbf{A} + \mathbf{B}$
- b. $\mathbf{A} - \mathbf{B}$
- c. $\mathbf{AB}$
- d. $\mathbf{BA}$

where

$$\mathbf{A} = \begin{pmatrix} 3 & 1 \\ 3 & 7 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 2 & 2 & 10 \\ 4 & 7 & 3 \end{pmatrix}$$

3. For each of the following, find the transpose in R and specify whether or not the matrix is symmetric using

`isSymmetric()` :

$$\mathbf{A} = \begin{pmatrix} 4 & 3 \\ 11 & 2 \end{pmatrix}$$

$$\mathbf{B} = \begin{pmatrix} 2 & 8 \\ 8 & 12 \end{pmatrix}$$

$$\mathbf{C} = \begin{pmatrix} 1 & 1 & 2 \\ 1 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$\mathbf{D} = \begin{pmatrix} 4 & 4 \\ 2 & 4 \end{pmatrix}$$

4. Using matrices **A** and **B** from question 3, calculate the following inverses in **R**:

- a. $\mathbf{A}^{-1}$
- b. $\mathbf{B}^{-1}$
- c. $(\mathbf{AB})^{-1}$
- d. $\mathbf{B}^{-1}\mathbf{A}^{-1}$ and verify that $(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$

5. Find the determinant of each of the following in **R**:

$$\mathbf{A} = \begin{pmatrix} 4 & 3 \\ 11 & 2 \end{pmatrix}$$

$$\mathbf{B} = \begin{pmatrix} 2 & 8 & 4 & 0 \\ 5 & 333 & 1 & 0 \\ 1 & 1 & 7 & 0 \\ 6 & 10 & 423 & 0 \end{pmatrix}$$

$$\mathbf{C} = \begin{pmatrix} 1 & 1 & 2 \\ 1 & 3 & 5 \\ 2 & 5 & 8 \end{pmatrix}$$

$$\mathbf{C}' \text{ and verify } |\mathbf{C}'| = |\mathbf{C}|$$

$$\mathbf{D} = \begin{pmatrix} 4 & 0 \\ 0 & 2 \end{pmatrix}$$

$$\mathbf{D}^{-1} \text{ and verify } |\mathbf{D}^{-1}| = 1/|\mathbf{D}|$$

6. Let the 4×4 matrix $\mathbf{A}$ be partitioned as

$$\mathbf{A} = \left(\begin{array}{ccc|c} 7 & 2 & 5 & 8 \\ -3 & 4 & 0 & 2 \\ 9 & 3 & 6 & 5 \\ 3 & 1 & 2 & 1 \end{array} \right) = \begin{pmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{22} \end{pmatrix}$$

In R, create the matrix $\mathbf{A}$ from above and partition it as shown. Assign appropriate values to $\mathbf{A}_{11}$, $\mathbf{A}_{12}$, $\mathbf{A}_{21}$ and $\mathbf{A}_{22}$.